ROLE 4: DATA AND RESILIENCE

A4 Documentation

1 Logical Data Model & Role/Access Rules

The STEMMate/STEMCraft data architecture is built on an offline-first paradigm tailored for Namibian schools characterized by shared, low-spec devices and intermittent or expensive internet connectivity. To guarantee data durability before synchronization occurs, all mutations are stored locally on the device.

Core Entities & Attributes:

- **Activities (`activities`):** Represents the cached activity catalog containing levels (primary, middle school, high school), topics, durations, and required materials.
- **Session Plans (`session_plans`):** Represents structured teaching plans created from selected activities. Attributes include `plan_id`(Primary Key), `user_id`, `title`, `target_group`, `sync_status`(`WAITING_TO_SYNC`, `SYNCING`, `SYNCED`, `FAILED`), `created_at`, and `updated_at`.
- **Drafts (`drafts`):** Stores temporary JSON-formatted records capturing user input dynamically via autosave (~0.7 seconds after change). Attributes include `plan_id`(Foreign Key / Primary Key), `payload_json` (validated via strict JSON constraints), and `updated_at`.
- **Outbox / Sync Queue (`outbox`):** Tracks local offline mutations awaiting synchronization in small batches (up to 20 items). Attributes include `queue_id`(Primary Key), `plan_id`(Foreign Key), `action`, `status`, `attempt_count`, and `created_at`.
- **Equipment Kit Custody (`equipment_kits`):** Managed under the custodian view to track kit checkouts, responsible facilitators, intended dates, and return statuses.

Entity Relationships & Database Constraints:

- **`session_plans` ↔ `drafts` (1-to-1 Relationship):** Each session plan links to its temporary working draft. This relationship enforces an `ON DELETE CASCADE`rule via `drafts.plan_id`, ensuring that deleting a session plan automatically purges its temporary draft to prevent orphaned records.
- **`session_plans` ↔ `outbox` (1-to-Many Relationship):** A session plan may generate multiple outbox mutation actions. This relationship uses an `ON DELETE SET NULL`rule on `outbox.plan_id`, preserving historical audit trails in the sync log even if a local plan entry is removed.
- **Automated State Synchronization:** A SQLite database trigger (`trg_sync_outbox_to_plan_update`) automatically keeps `session_plans.sync_status`aligned with outbox status updates.

Role and Access Governance Rules:

- **Facilitator View:** Authorized to browse the local activity catalog (`FR1`, `FR2`), save activities offline, create and edit structured session plans with automated local autosaves (`FR3`, `FR4`, `FR6`), and trigger outbox synchronization (`FR5`).
- **Equipment Custodian View:** Authorized to access the Equipment Kit Custody interface to log, edit, or review kit checkouts and return statuses (`FR7`).
- **Privacy Guardrails:** Authentication and sign-in are intentionally out of scope to prevent collecting sensitive personal identification data, keeping the design focused strictly on role-based view switching with synthetic data only.

2 Offline & Sync Strategy

Based on workflow **W3** and the team's resilience design:

- **Local-First Transaction Flow:** Every user action (saving an activity, updating a plan draft, or finalizing a record) writes directly to the local device storage first, ensuring work continues smoothly offline.
- **Lifecycle State Machine:**
    1. *Save:* Work is stored locally and queued in the outbox.
    2. *"Waiting to sync":* Displayed in the status bar with a pending item count when offline.
    3. *Syncing:* Triggered automatically upon network reconnection or manually via the **Sync** button in small batches of up to 20 items.
    4. *Confirmed Result:* The status bar updates to *"All changes synced"* once the server layer acknowledges the batch.

3 Durable SQLite Schema (Patched & Production-Ready)

This incorporates the patched SQLite schema definitions from the repository fixes (addressing foreign keys, JSON validation, and status triggers):

```
-- Session Plans table
CREATE TABLE IF NOT EXISTS session_plans (
    plan_id TEXT PRIMARY KEY,
    user_id TEXT,
    title TEXT NOT NULL,
    target_group TEXT NOT NULL,
    sync_status  TEXT  DEFAULT  'WAITING_TO_SYNC',
    created_at TEXT NOT NULL DEFAULT (STRFTIME('%Y-%m-%d %H:%M:%f', 'NOW')),
    updated_at TEXT NOT NULL DEFAULT (STRFTIME('%Y-%m-%d %H:%M:%f', 'NOW'))
);

-- Drafts table with JSON validation and cascade delete
CREATE TABLE IF NOT EXISTS drafts (
    plan_id TEXT PRIMARY KEY REFERENCES session_plans (plan_id) ON DELETE CASCADE,
    payload_json TEXT NOT NULL CHECK (json_valid(payload_json)),
    updated_at TEXT NOT NULL DEFAULT (STRFTIME('%Y-%m-%d %H:%M:%f', 'NOW'))
);

-- Outbox table for offline mutations with indexed status
CREATE TABLE IF NOT EXISTS outbox (
    queue_id INTEGER PRIMARY KEY AUTOINCREMENT,
    plan_id TEXT REFERENCES session_plans (plan_id) ON DELETE SET NULL,
    action TEXT NOT NULL,
    status TEXT NOT NULL DEFAULT 'WAITING_TO_SYNC'  CHECK (status IN ('WAITING_TO_SYNC',
'SYNCING', 'SYNCED', 'FAILED')),
    attempt_count INTEGER NOT NULL DEFAULT 0 CHECK (attempt_count >= 0),
    created_at TEXT NOT NULL DEFAULT (STRFTIME('%Y-%m-%d %H:%M:%f', 'NOW'))
);

CREATE INDEX IF NOT EXISTS idx_outbox_status ON outbox (status);

-- Trigger to keep plan sync_status in sync with outbox changes automatically
CREATE TRIGGER IF NOT EXISTS trg_sync_outbox_to_plan_update
AFTER UPDATE OF status ON outbox
BEGIN
    UPDATE session_plans
    SET sync_status = NEW.status,
        updated_at = STRFTIME('%Y-%m-%d %H:%M:%f', 'NOW')
    WHERE plan_id = NEW.plan_id;
END;
```

4. Architecture Decision Record (ADR 3)

- **Status:** Accepted (Carried forward from the A3 resilience design)
- **Context:** STEMCraft operates in Namibian schools characterized by patchy, expensive internet connections and shared, low-spec hardware. The system must ensure zero data loss during sudden connection drops.
- **Decision:** Implement a local-first SQLite database architecture combined with an outbox queue pattern. Mutations are stored locally immediately, and synchronization occurs in controlled batches of up to 20 items via explicit user or automatic reconnection triggers.
- **Consequences:** User workflows are never blocked by a lack of network access; however, data persists locally on shared laptops, requiring future consideration for data purging or deletion mechanisms once records are fully confirmed by the remote server.